Forcing Reading Group — Exercise Sheet 3

The Paris–Wilkie Forcing

Remark 1. In this section we will fix a countable nonstandard $M \models \mathrm{Th}(\mathbb{N})$ and $n \in M \setminus \mathbb{N}$.

Exercise 2. Let $PHP(f)$ be the formula

$$(\forall x)((\exists y \leq x)(f(y) \geq x) \vee (\exists y_1 \leq x)(\exists y_2 \leq x)(y_1 \neq y_2 \wedge f(y_1) = f(y_2))).$$

Try to explicitly define a function $f : \{0, \ldots, n+1\} \to \{0, \ldots, n\}$ such that $(M, f) \models \neg PHP(f)$. Does your $(M, f) \models I\exists_1(f)$?

Definition 3. Let P be the set of all (standardly) finite partial injections from $\{0, \ldots, n+1\}$ to $\{0, \ldots, n\}$. For all $p, q \in P$ we define $q \leq p \Leftrightarrow q \supseteq p$.

Exercise 4. Every (standardly) finite set of pairs of elements of M can be represented by a single element of M by a suitable encoding. Show that P, interpreted as a set of codes, is not definable in M.

Remark 5. Our goal is to build a more sophisticated f such that $(M, f) \models I\exists_1(f) + \neg PHP(f)$. We will build f as a union of a chain $p_0 \geq p_1 \geq p_2 \geq \ldots$ in P. We will pick the p_i's such that:

- Every pigeon $y \in \{0, \ldots, n+1\}$ is mapped somewhere.
- Every $\exists_1(f)$-formula φ satisfies $LNP(\varphi)$.

Exercise 6 (Locality of quantifier free formulas)**.** Show that if $(M, f) \models \varphi$, where f is total and φ is quantifier free $L_{PA}(f)$-formula with parameters from M, then for any $p \in P$ such that $p \subseteq f$, there is $p \subseteq q \subseteq f$ such that for all $f' \supseteq q$: $(M, f') \models \varphi$.

Exercise 7. Define a relation $\Vdash$ between P and quantifier free $L_{PA}(f)$-formulas with parameters from M by induction on the complexity of the formulas such that: $p \Vdash \varphi \Leftrightarrow \forall f \supseteq p : (M, f) \models \varphi$.

Exercise 8. Extend the previous relation to $\exists_1(f)$-formulas with parameters from M with the condition weakened to $p \Vdash (\exists x)\varphi(x) \Rightarrow \forall f \supseteq p : (M, f) \models p$.

Moreover, make sure that for all $p \in P$:

$$(\exists q \leq p)(q \Vdash (\exists x)\varphi(x)) \Leftrightarrow (\exists f \supseteq q)(M, f) \models (\exists x)\varphi(x).$$

Exercise 9. Show that for every $\exists_1(f)$-formula φ and every $p \in P$, the set

$$\{a; \exists q \leq p : q \Vdash \varphi(a)\}$$

is definable in M.

Exercise 10. Construct an f such that $(M, f) \models I\exists_1(f) + \neg PHP(f)$.